CROPPING SYSTEM

LAST 5 YEARS (2019-2023) PREVIOUS YEAR QUESTIONS FROM CROPPING SYSTEM

1. **Which of the following statement is false about mixed cropping.**

   a. Growing of crop in fixed row & pattern is called as mixed cropping.
   b. Mixed cropping generally practiced in dry region.
   c. Seed of both crops are sown by broadcasting.
   d. Mixed cropping is generally substantial in nature.

2. **Identify the type of cropping system in which crop is harvested from stem or above the ground level & left to regrow for the next season.**

   a. Relay cropping
   b. Sequence cropping
   c. Ratoon cropping
   d. Alley cropping

3. **The type of cropping system in which different crops are sown in definite row & pattern**

   a. Mixed farming
   b. Mixed cropping
   c. Intercropping
   d. Crop rotation

4. **Growing of short duration crop between the rows of a sugarcane is an example of which type of cropping system**

   a. Intercropping
   b. Mixed cropping
   c. Alley cropping
   d. Double cropping

5. **Which cropping system crops has the maximum area under cultivation in India.**

   a. Cotton & groundnut-based cropping system
   b. Rice & wheat-based cropping system
   c. Maize & legume-based cropping system
   d. None of the above

6. ---------------refers to the intensification of cropping in sequence on the same field per year, the succeeding crop is planted before the preceding crop has been harvested, crop intensification is only in the time dimension, there is no intercrop competition.

a. Relay cropping
b. Sequence cropping
c. Companion cropping
d. Ratoon cropping

7. ____________refers to the growing of two crops or more crops in a proper sequence on a piece of a land in a given time period.

a. Companion cropping
b. Sequence cropping
c. Crop rotation
d. Mixed cropping

8. What is the farming system that uses natural pest controls and biofertilizer and in which the use of any type chemical and synthetic pesticides, fertilizer is completely avoided?

a. Contingency cropping
b. Companion cropping
c. Double farming
d. Complimentary cropping

9. It is the type of farming system in which livestock, crops, poultry are raised on a single field for the purpose of subsistence by the farmer is referred as

a. Mixed farming
b. Mixed cropping
c. Cropping system
d. Complimentary cropping

ANSWERS

1. A 2. C 3. C 4.A 5.B 6.A 7.B 8.B 9.A

CROPPING SYSTEM

System approach is **applied in agriculture for efficient utilization** of all resources, maintaining stability in production & obtaining higher net returns.

FARMING SYSTEM

Farming system is defined as a set of various elements or components that are interrelated among themselves. Farming system consist of several enterprises like **cropping system, dairying, piggery, poultry, fishery, bee keeping etc.**

Characteristics Of Farming System-

1. All **the components are interrelated** among themselves.
2. The **end product or waste of one enterprise (component)** is used as an input in other enterprise.
3. **Main aim of farming system is increasing profitability** to the farmer & sustainability.
4. All the component **interacts among themselves & with the environment** without altering ecological & socio-economic balance.

MIXED FARMING

It is the system of farming on a particular farm which include **crop production, raising livestock, poultry, fisheries, beekeeping** Etc., **to sustain & satisfy as many** needs of the farmer as possible.

Image (i) mixed farming

CROPPING SYSTEM

Cropping system means **the proportion of an area under various crops at a point of time in a unit area**. It indicates the **yearly sequence & spatial arrangement** of crops and fallow in an area.

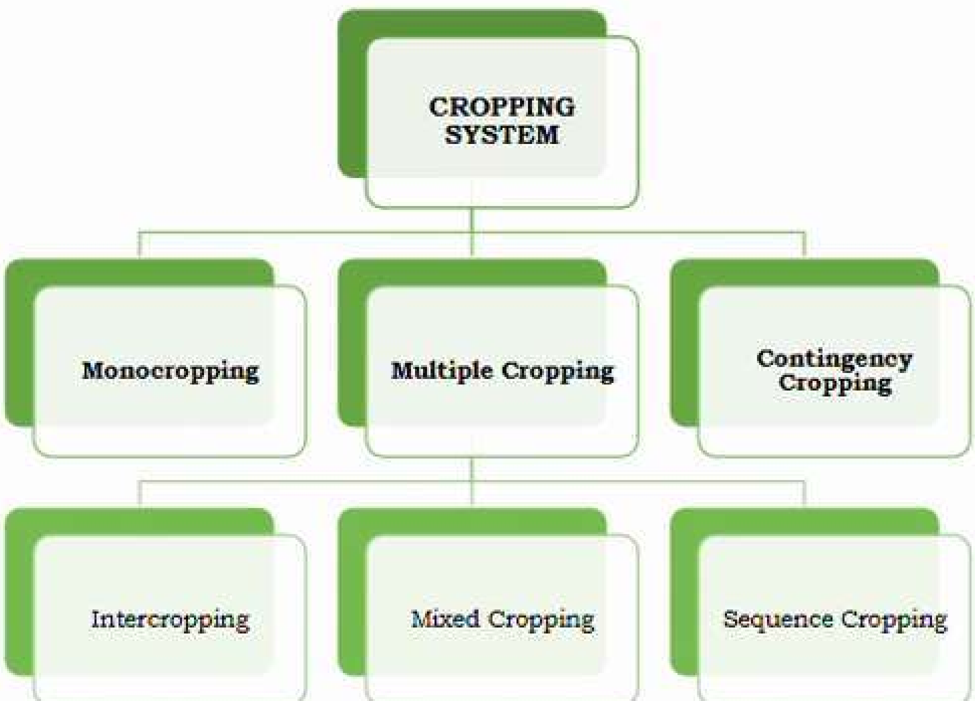
Characteristics:

1. Important **component of farming system**.
2. Interact with all the **available farm resources ex soil, water, microbes**, environment Etc. & all the available technology.

Ex- wheat, rice, cotton, groundnut-based cropping system.

TYPES OF CROPPING SYSTEM

Cropping system **is classified into 3 types based** on available resources & technology.



1. MONOCROPPING –

Monocropping refers to the **growing of only one crop on a piece of land.** It may be due **to climatological & socio-economic conditions or** due to specialization of a farmer in growing a particular crop.

Ex- 1**. Rainfed condition-** groundnut, sorghum or cotton

i. **Specialization**-flue-cured tobacco in Guntur region of Andhra Pradesh

Image (ii) – maize as a monocrop

Advantages Of Monocropping

- ✓ Convenience in **sowing with the help of machinery** under mechanized farming.
- ✓ It is **convenient for harvesting** with the help of machinery.

Disadvantages of monocropping

- ✓ Reduce **fertility & productivity** of the soil.
- ✓ **Cause degradation of the soil structure**.
- ✓ **Impact biodiversity of that** region.
- ✓ Increase **pest & weed attack**.
- ✓ Increase **the pest & weed resistance**.

2. MULTIPLE CROPPING

Growing of **two or more crops on the same piece of land in** one year calendar is known as multiple cropping.

It is **the intensification of the cropping in time & space dimension** means getting a greater number of crops from a single piece of land in a given time period.

Types Of Multiple Cropping

1. Intercropping
2. Mixed cropping
3. Sequence cropping

INTERCROPPING

Growing of two or more crops simultaneously on the same piece of land in a definite row & pattern.

Ex Seteria + red gram (5 :1 ration pattern)

Types of intercropping-

Based on number of plant population intercropping is divided into two parts.

Image (iii) intercropping of cabbage + ginger in definite row

1. **Additive series.**
2. **Replacement series.**

Additive series-in additive series we **maintain the 100 & population of our base crop** (main crop) but changes the geometry of the main crop to adjust the intercrop in between.

Ex wheat & mustard in 10:1(means 10 rows of wheat with 1 row of mustard)

In additive series the population of the intercrop is very less.

Replacement series-In replacement series the population **of the one crop is reduced to adjust the intercrop in between.** Here both the crops are called as a component crop

Ex-castor & red gram grown in 1:1

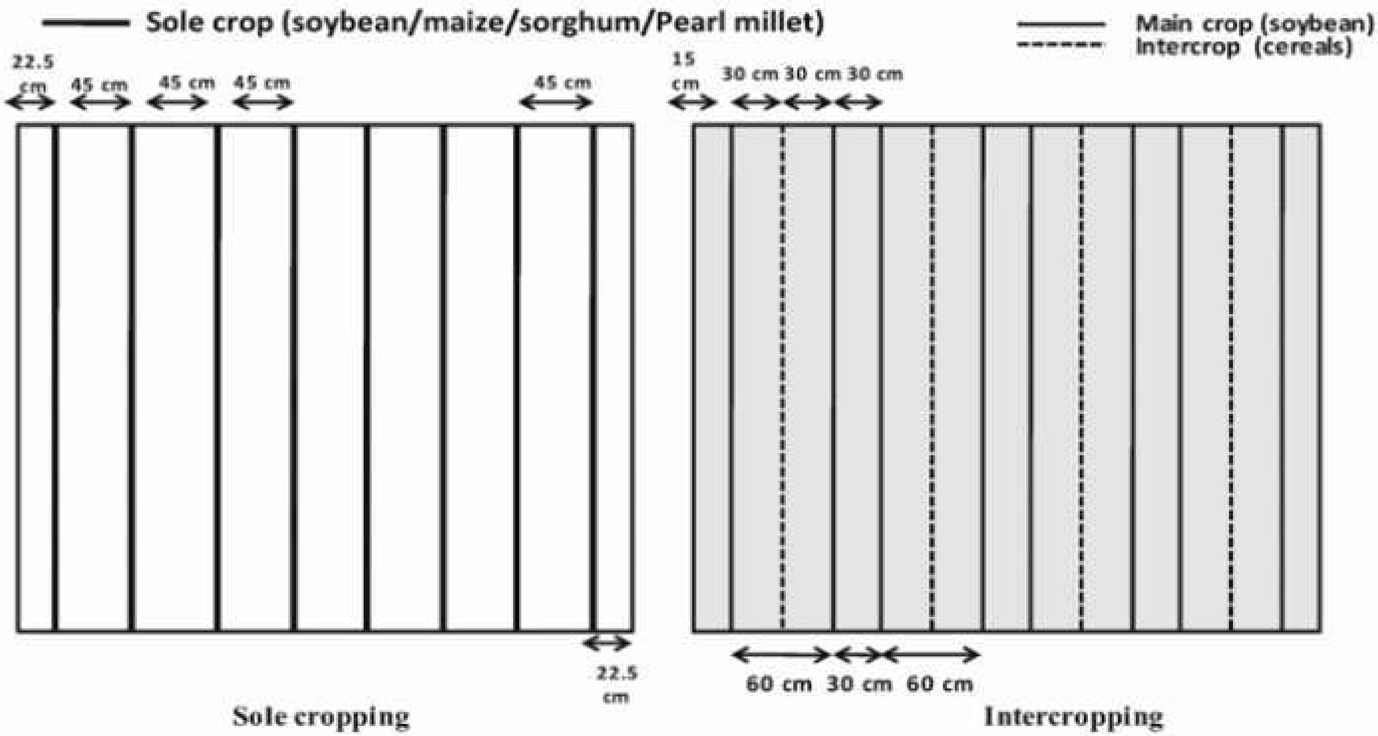


Image (iv) - Geometry change in intercrop

Intercropping may be divided in following four groups: based on arrangement of crops in a field.

1. Parallel cropping

Under this **cropping two crops are selected which have** different growth habits and have a zero competition **between each other and both of** them express their full yield potential.

Example: Mung or urd with maize and urd, mung or soybean with cotton.

Image (v) – Parallel Row Intercropping of Maize + Green moong

2. Companion cropping

In companion **cropping the yield of one crop is not affected by the other**. In other words, **the yield of both the crops is equal** to their pure crop. Thus, the standard plant populations of both crops are maintained.

Example: Mustard, wheat, potato, etc. with sugarcane

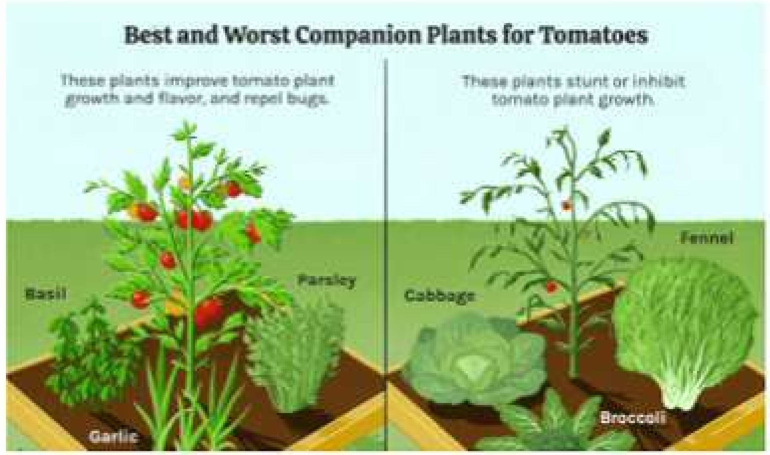


Image (vi) – companion crops for tomato (basil, garlic, parsley)

3. Multi-storeyed cropping

Growing plants of different heights in the same field at the same time is termed as multi-storeyed cropping. It is mostly practiced in orchards and plantation crops for maximum use of solar energy even under normal planting density.

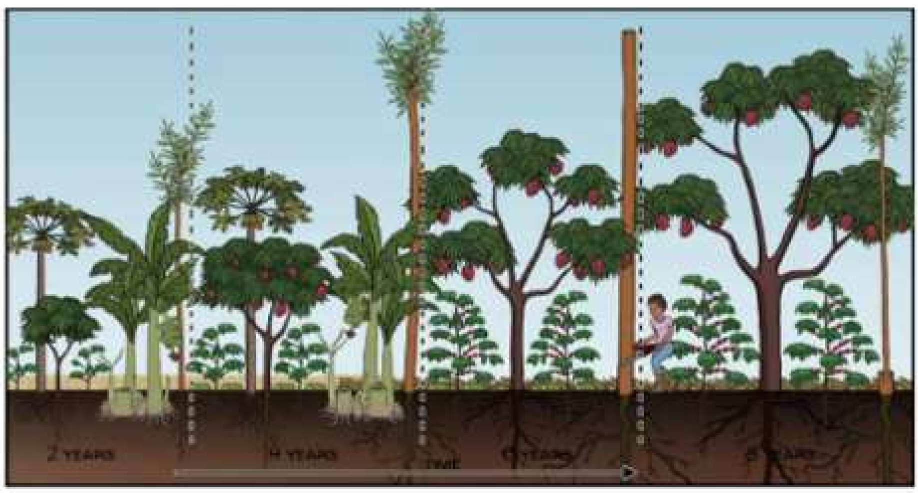


Image (vii)- Multi-Story Cropping

Example: Eucalyptus, papaya, and berseem grown together, **Growing of pineapple, sweet potato, black pepper, tapioca, turmeric, ginger etc. in coconut** or areca-nut.

Sometimes it is practiced under field crops such as sugarcane and potato and onion (seed crop) or sugarcane, mustard and potato.

Advantages of Multi-Storey Cropping System

- ✓ It gives **maximum production from small plots**. This can help farmers cope with land shortages along **with income per unit area increase** substantially.
- ✓ Minimizes the **risk of crop yield loss**.

- ✓ Improves **physical properties and health** status of the soil.
- ✓ Including legumes in the **cropping pattern helps maintain soil fertility** by nitrogen fixation in the soil.
- ✓ Different **types of crops can be produced thereby** providing a balanced diet for the family.
- ✓ Weeds are **suppressed due to high density planting**.
- ✓ Saves the crop from climatic **aberrations like high rainfall, soil erosion, landslides** etc.
- ✓ Maintain **an ecological balance.**
- ✓ Provides suitable **micro-climate conditions that benefits** the winter crops.
- ✓ Efficient use **of resources available**.
- ✓ **Helps in maintaining ecological balance**

Challenges In Adopting:

- ✓ **Drought conditions**.
- ✓ Lack of funds.
- ✓ Lack of technical **knowledge of cropping systems**
- ✓ Timely availability of inputs.
- ✓ **Pest and disease** incidents.
- ✓ Lack of **irrigation facilities.**
- ✓ **Lack of labour** availability.

4. Synergistic cropping

Here the **yields of both crops, grown together are found to be higher** than the yields of their pure crops on unit area basis.

Example: Sugarcane and potato

Image (viii)- synergetic cropping of sugarcane + potato

Reasons to practice intercropping

1. As an insurance **against the failure of one crop** in rainfed condition.
2. To get **higher productivity** per unit area.
3. **To get stability** in production.
4. For **efficient utilization** of resources.

Principles of inter cropping

- ✓ The crops grown in association should have ***complementary effects*** rather competitive effects.
- ✓ The subsidiary crop should be of ***shorter duration and of faster growing habits*** to utilize the early slow growing period of main crop and they must be harvested when main crop starts growing
- ✓ Ex-Autumn planted **sugarcane remains dormant after germination uptil February** during which **potato, berseem, lucerne, mustard, etc.** could be taken successfully as companion intercrops.
- ✓ The component crops should have ***similar agronomic practices***.
- ✓ Erect growing crops should **be intercropped with cover crops like pulses** so that the soil erosion and weed population could be reduced or checked.
- ✓ The ***component crops should have different root depths*** so that they do not compete for nutrients, water and root respiration among them. Ex one should have shallow root & other should have deeper root.
- ✓ A ***standard plant population of main crop should be maintained*** whereas that of subsidiary crops the plant population could be increased or decreased as per demand of the situation.
- ✓ Component crops of similar pest and disease pathogens and parasite infestations should not be chosen.
- ✓ The ***planting method and management should be simple***, less time taking, less cumbersome, economical and profitable so that it may have wider adoptability.

Example: Maize intercropped with green gram, black gram or groundnut etc.

Requirements For Intercropping

1. The time for ***peak nutrient demand*** for main & Intercrop should not overlap.
2. Very ***less competition*** for the light between the crops.
3. **Both the crops be it main crop or inter crop** should be complimentary in nature.

4. The ***difference of maturity*** between intercrops should be at **least 30 days**.

Examples of intercrop

- ✓ **Groundnut + pigeon pea**- in 11:1 row ratio.
- ✓ **Sorghum + pigeon pea** in 2:1 row ratio.
- ✓ **Groundnut + castor** in 4:1 row ratio.
- ✓ **Chickpea+ safflower** in 3:1 row ratio.
- ✓ **Finger millet + pigeon pea** 8 :1 row ratio.

Advantages Of Inter-Cropping

- ✓ The nutrients from **different layers of the soil are evenly** used. A cereal-legume mixture is beneficial because **of an efficient fixation of atmospheric nitrogen** into the soil.
- ✓ Total **bio-mass production/unit area/period of time** is increased because of the fullest use of land as **the inter-row space will be utilized** which otherwise would have been used for weed growth. Thus, the profit/unit area becomes high.
- ✓ The **fodder value in terms of quantity and quality becomes higher** when a non-legume is intercropped with legume **viz. Napier + cowpea, Napier + berseem.**
- ✓ It **provides crop yields in instalments** which reduces the marketing risks.
- ✓ It offers best **employment and utilization of labour, machine and power** throughout year.

Disadvantages Of Intercropping

- ✓ **Fertilizer management is difficult** because the nutrient requirement of the crops is different,
- ✓ Difficulty in **harvesting because of different seeding time of crops**.
- ✓ There **are certain combinations which suppress the growth of another crop** and may be conducive to insect pests and diseases.

MIXED CROPPING

Growing of two or more crops simultaneously intermingled **without any fix row & pattern.**

Characteristics

1. Seeds of different **crops are mixed in a proportion** & sown.
2. Mostly **practiced in dry land regions** of India.
3. Main **objective is to meet the** family requirement of vegetables, cereals, pulses etc.
4. Mixed **cropping is of subsistence** in nature.

Image (ix)- mixed cropping of maize + pumpkin without a specific row & pattern

Principles Of Mixed Cropping

The following points **should be considered while selecting crops**.

- ✓ **Legumes should be sown with non-legumes** examples arhar with jowar, gram with wheat.
- ✓ **Tall growing crops should be sown with short** growing crops. e.g. maize with mung/urd.
- ✓ **Deep rooted crops (tap rooted crops) should be sown with shallow** rooted or adventitious crops.
- ✓ **Bushy crops should be sown with erect** growing crops.
- ✓ Crops being **attacked by similar insects, pests and diseases** should not be sown together.
- ✓ **Mixture should consist short and long duration crops**.

Advantages

- ✓ Risk of **failure of crop is less**
- ✓ Fulfils the **daily requirements** of food grains, oilseeds, pulses etc.
- ✓ **Improve fertility of the soil if legumes** are taken as minor crop
- ✓ Better **distribution of labour throughout** the crop period
- ✓ **Increase gross monetary** returns
- ✓ Well **balanced cattle feed** is obtained
- ✓ **Safeguards against pests** and diseases
- ✓ Full utilization of **space and available plant** nutrients

Disadvantages

- ✓ Sometimes **control of pests, diseases and weeds** become difficult
- ✓ Sometimes affects **the yield of main crop**
- ✓ **Harvesting with the help of** machinery is not possible

SEQUENCE CROPPING

It is defined as **growing of two or more crops in a sequence** on the same piece of land in a farming year.

Depending upon the number of crops grown in a field in a year it is called as **double (if 2 crops), triple (if 3 crops) quadruple etc**.

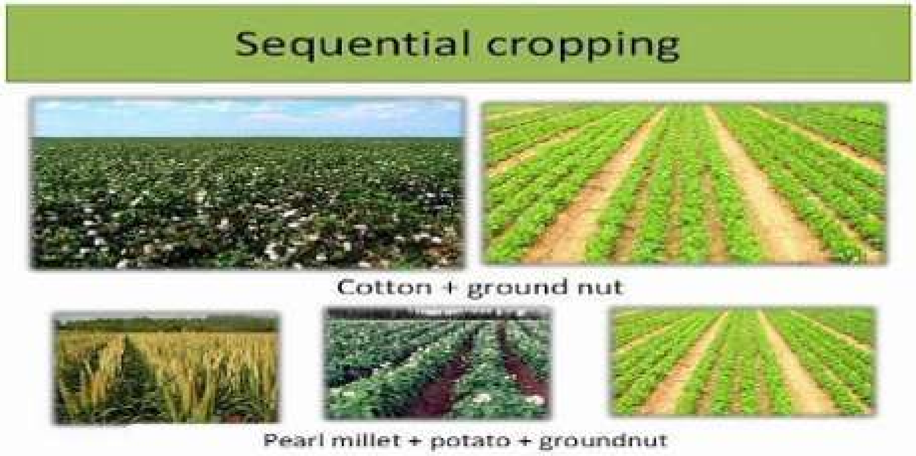


Image (x)- sequence cropping i- 1st season - coconut & 2nd season groundnut ii- 1st season pearl millet followed by potato & in 3rd season groundnut

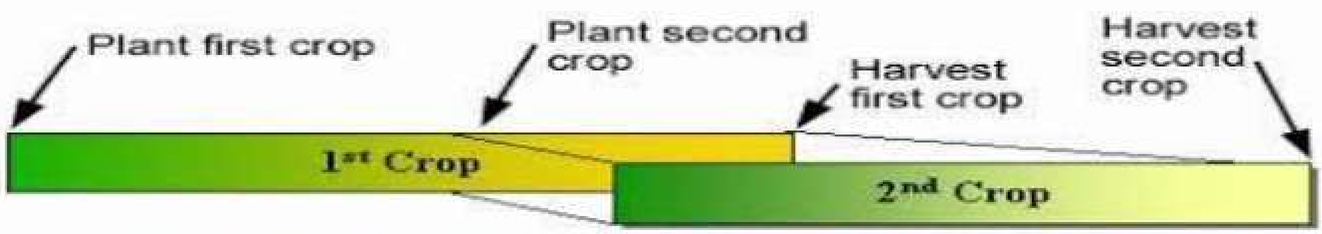
Example of double cropping- rice-linseed, rice-vegetables, soyabean-wheat etc.

Double cropping or ***more is feasible in areas of rainfall more than 850 mm.***

Other Types of Sequence Cropping Are

1. Relay cropping.
2. Ratoon cropping.
3. Alley cropping

Relay cropping-planting of the succeeding crop before the harvesting of the preceding crop. **Ex Paddy-Lathyrus, Paddy-Lucerne, Paddy-Berseem, Cotton-Berseem**



Advantages:

Image (xii) Soyabean is sown before harvesting of wheat crop

- ✓ **Better utilization of residual moisture** and fertilizers.
- ✓ Reduces the **cost of cultivation** practices.
- ✓ Also reduces **the cost of fertilizers** and irrigation.
- ✓ **Labour requirement** is less.
- ✓ Incidence of **pest, diseases and weeds** is less due to early sowing operation.

Disadvantages:

- ✓ Risk of **crop failure** is more.
- ✓ Harvesting by means **machinery is difficult**.
- ✓ Lack of **availability of skilled labour**.
- ✓ **Greater incidence of pest**, disease and weeds.

Ratoon cropping-

To raise a crop with regrowth **coming out of roots or stalks after the** harvest of the crops. **Ratooning is one of the important systems of intensive cropping, which implies more than one harvest from one sowing/planting because** of regrowth from the basal buds on the stem after harvest of first crop. Thus, ratooning consists of allowing the stubbles of the original crop to strike again or to produce the tillers after harvesting and to raise another crop.**e.g. Ratooning of Sugarcane, Hybrid Jowar, Hybrid Bajra, and Redgram etc.**

Image (xiii) Ratooning of sugarcane

Alley cropping-

The system of growing jowar, maize, bajra or any other arable crop in the alleys (passage between two rows) of leguminous shrubs like subabul (Leucaena leucacephala) is called as alley cropping.

Image (xiv)- maize is grown as fodder between alley of trees

Types of Alley cropping: -

1. **Food cum fodder system**: In this system provides food grains like pulses, cereals, oilseeds and fodder for livestock.

2. **Food cum mulch system**: In this system **provides food grains as well as crop residues** as a mulch for soil and water conservative measures.

3. **Food cum pole system**: In this system **provides food as well as wood for** fuel, timber, furniture etc.

Advantages:

- ✓ Better **utilization of natural** resources
- ✓ **Reduces the cost** of cultivation.
- ✓ Improves **the soil fertility and productivity**.
- ✓ **Incidence of pest**, diseases
- ✓ Provides **fodder for animals** and food for human.

Disadvantages

- ✓ Competition among the **natural resources i.e., moisture, nutrients**, light and space.
- ✓ Incidents of **pest, diseases and weeds** is more
- ✓ Chances **of crop failure** is more
- ✓ **Less yield** is obtained

CONTINGENCY CROPPING

It is defined as a **growing of a suitable crop in place of normally grown crop** in a region due to late receipt of rainfall.

Generally contingency **crops give low yields** as compared to **normal grown crop**.

Image (xv)-gap filling with pear millet due to failure of main crop

COMPANION CROPPING

Companion cropping is a type of polyculture under **which two plant species grown together which synergistically improve one another's growth**. These crops directly mask the **negatives of each other & enhance the positive impacts** of each other. The push & pull technology to **reduce weed & pest** is based on companion cropping.

Image (xvi)-companion cropping of onion + carrot as onion scent repel insect from carrot crop

Companion cropping includes the following

1. Trap crop
2. Repellent crop
3. Masking plant
4. Camouflage crop

Trap crop- trap **crops are the plants that attract the pest away from the target crop. Once** the trap crop concentrated with pest then pest can be removed by **burning, tilling, insecticide, etc.** or can be left untouched.

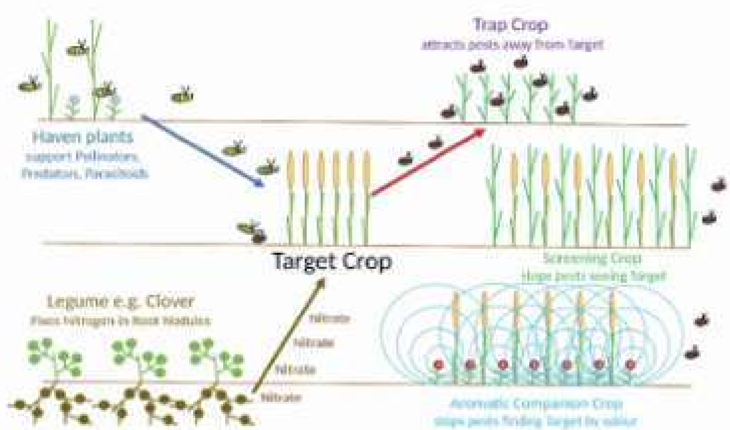


Image (xvii)-Trap crop attracts insects from main crop

Ex- Napier grasses a trap crop for stem borer & marigold attract thrips from tomato crops.

Repellent crop

Certain plant **releases some chemical compound which repel** or deter insect pest from the main crop.

Ex- basil planted with tomato reduce tomato thrips Onion repels insects like arthropods, mites, aphids etc.

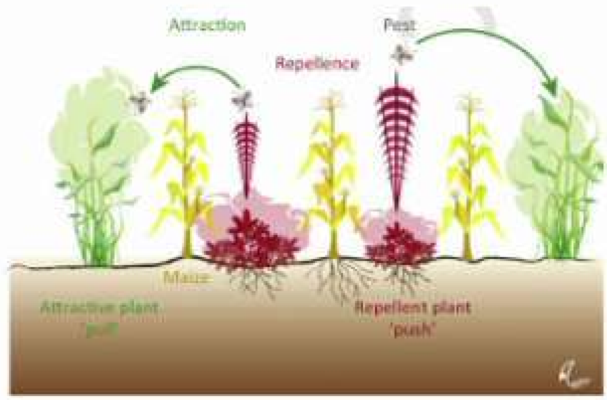


Image (xviii)-repellent crop (push) repel pest from the main crop & attractive plant attract (pull) pest from main crop

Masking Plants-

These plants release **some volatiles that masks host plant odors** interfering with host plant location.

Ex - cabbage root fly was disrupted when **peas, rye-grass or clover** planted on its broader.

Image (xix)-Rye grass & clover release odor to mask the host plant from pest

Camouflage Plants-

Are those plant which **physically & visually** blocks the host plant. **Ex - intercropping** of cabbage reduce pest load on cabbage.

Image (xx)- cabbage gets camouflaged by the intercrop

CROP ROTATION

Crop rotation is a process of ***growing different crops in succession on a piece of land*** in a specific period of **time, with an object to get maximum profit from least** investment without impairing the soil fertility.

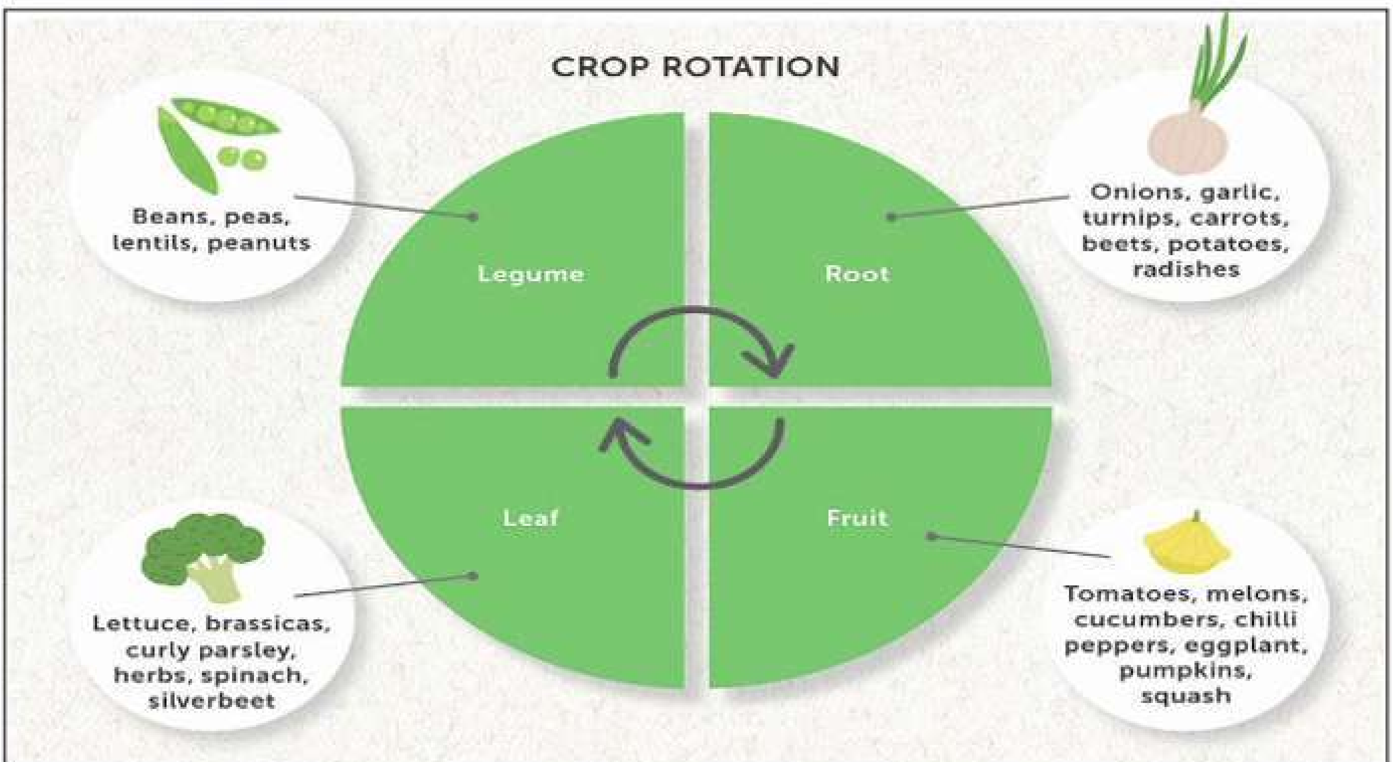


Image (xxi) crop rotation (1st season root crop, 2nd season fruit (melon), 3rd season leaf crop & 4th season legume crop)

PRINCIPLES OF CROP ROTATION

- There are **certain accepted principles based** on which the crops should be selected for crop rotation
- The crops with tap roots should be followed by **those which have fibrous root system**.
- The **leguminous crops should be grown after non-leguminous** crops because legumes fix atmospheric nitrogen into the soil and add more organic matter to the soil; while on the other hand, non-legumes are fertility depleting crops.
- **More exhaustive crop should be followed by less exhaustive crops** because crops like potato, sugarcane, maize etc. need more inputs such as better tillage, more fertilizers, greater number of irrigations, more insecticides, pesticides etc.
- **Selection of the crops should be demand based** i.e. the crops which are needed by the people of the area & can be easily sold.
- **The selection of crops should be problem based** e.g. On sloppy lands which are prone to soil erosion, an alternate cropping of erosion promoting (erect growing crops like millet, etc.) and erosion resisting crops like legumes, should be adopted.
- **Under dry farming or partially irrigated areas the selection** of crops should be such which can tolerate the drought similarly in low lying and flood prone areas the crops should be such which can tolerate **water stagnation e.g. paddy, jute etc**.
- The selection of **crops should suit the farmers' financial conditions**.
- The crops **selected should also suit the soil and climatic conditions**.
- **The crops of the same family should not be grown** in succession because they act like alternate hosts for insects, pests and disease pathogens. **Ex Johnson grass grow with gramineous crop throughout year.**

- **<u>An ideal crop rotation is one which provides maximum employment</u>** to the family and farm labour, the **machines and equipment's are efficiently** used and all the agricultural operations are done timely.

ADVANTAGES OF CROP ROTATION

- Agricultural operations can **be done timely for all the crops because** of less competition. The supervisory work also becomes easier.
- Soil fertility is **restored by fixing atmospheric nitrogen, encouraging microbial activity, avoiding accumulation of toxins (HCN etc.)** and maintaining physic-chemical properties of the soil.
- The soil may also be **protected from erosion, salinity and acidity**.
- An ideal crop rotation helps **in controlling insects, pests and diseases. It also controls the weeds in the fields** e.g. Phalaris minor in wheat, Kasni weed in berseem.
- **Proper utilization of all resources and inputs** could be made by following crop rotation.
- The farmer **gets a better price for his produce because of its higher demand** in the locality or market.
- **Growing crops of different root depths avoids** continues depletion of nutrients from same depths e.g. the deep-rooted crops take nutrients from deeper zone and during that period the upper zone gets enriched.
- Best utilization of **residual moisture, fertility and organic residues is** made by growing crop of different nature.
- Ideal crop **rotation improves percolation, soil structure and reduces** chances or creation of hard pan in sub-soil zone.
- **Some crop plants are found to produce phytoalexins** when they get infected by diseases. Repeated cultivation of such crops results in harmful effects over crop plants and lower crop yield is obtained.

INTERACTION BETWEEN CROPS IN A CROPPING SYSTEM

Be it **mixed cropping or intercropping plant interact among themselves** for various resources which are required for the **plant growth such as solar radiation, water, nutrients, carbon dioxide etc.**

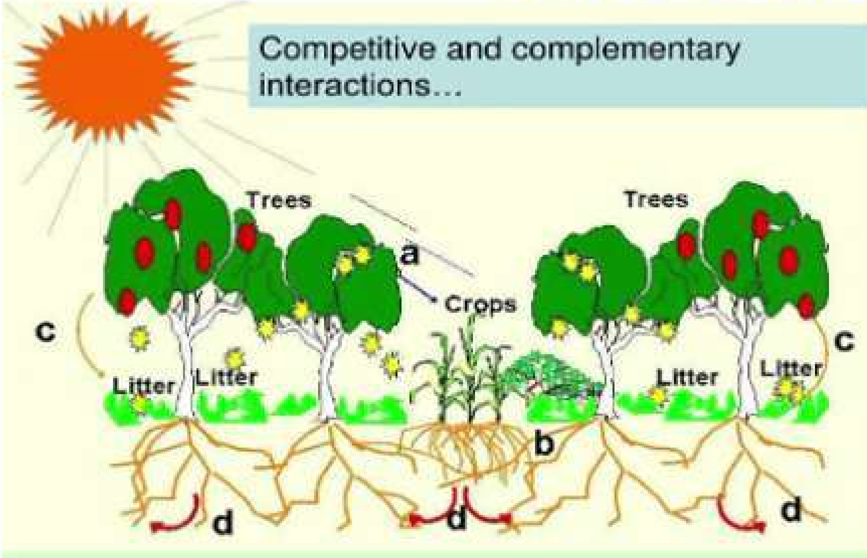


Image (xxii)- competitive & complimentary interaction between different crop

Allelopathy:

It is ***direct or indirect harmful effect*** that one plant has on another plant by the release of chemical substance or toxin that hamper the root growth of the other plant

Ex juglone released by walnut plant which suppress the growth of other plant in that area.

Annidation

it is the ***complimentary interaction*** which occurs between plants **for both space & time**. These crops have complimentary impact on each other's growth

Ex Seteria + red gram. They have different peak nutrient demand, different harvesting days, different root system.

4. Other complimentary effect

- ✓ Erect crop act as a support to the **climber ex coconut + pepper, maize + beans.**
- ✓ Taller crops act as a wind barrier **crop & protect the short crop ex maize + groundnut, onion+ castor, & turmeric + castor.**
- ✓ **Legume crop cause mobilization of the nitrogen in the rhizosphere** of the other crop or for succeeding crop as well.

Some important facts about cropping system

1. **Sorghum & pearl millet stubbles have high C:N ratio** which cause immobilization of the nitrogen & deficiency of the N_2 in the early stages of the succeeding crop.

2. **Legumes crops & their residue** add nitrogen in the soil.

3. **Guard crops**: under this system of cropping, the **main crop is grown in the centre, surrounded by hardy or thorny crops such as safflower around pea**, or wheat, Mesta (Patsan) around sugarcane, jowar around maize etc. With a view to provide protection to the main crop.

4. **Augmenting crops**: when sub **crops are sown to supplement the yield of the main crop the sub crops are called as Augmenting crops such as Japanese** mustard with berseem. Here the mustard helps in getting higher tonnage of fodder inspite of the fact that berseem gives poor yield in first cutting.